# ECE250: Signals & Systems Quiz-3

**Date:** 10-September-2025 **Total Points:** 10 **Duration:** 15 Minutes

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**Note:**

1. Please provide proper mathematical justifications with your answers. No marks will be awarded without a valid justification.
2. Institute Plagiarism policy are strictly applicable.

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[CO1] **[Q1]**: Let

$$x(t) = u(t-3) - u(t-5) \text{ and } h(t) = e^{-3t}u(t)$$

(a) Compute $y(t) = x(t) * h(t)$. **[4 Points]**

(b) Compute $g(t) = (\frac{dx(t)}{dt}) * h(t)$. **[4 Points]**

(c) How is $g(t)$ related to $y(t)$? **[2 Points]**